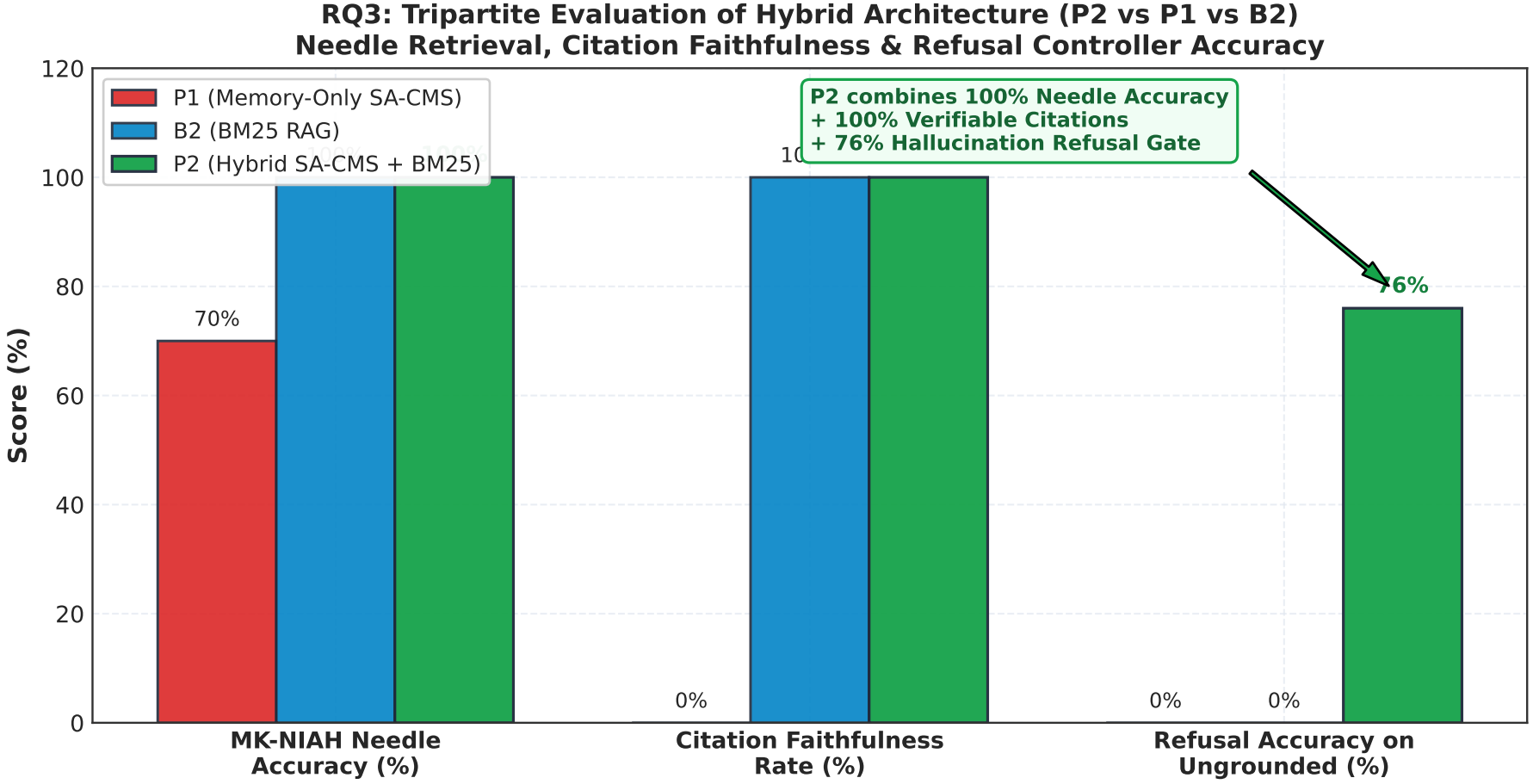
RQ3: Tripartite Evaluation of Hybrid Architecture (P2 vs P1 vs B2)
Needle Retrieval, Citation Faithfulness & Refusal Controller Accuracy
P1 (Memory-Only SA-CMS)
B2 (BM25 RAG)
P2 (Hybrid SA-CMS + BM25)
P2 combines 100% Needle Accuracy
+ 100% Verifiable Citations
+ 76% Hallucination Refusal Gate
Score (%)
120
100
80
60
40
20
0
70%
0%
76%
MK-NIAH Needle Accuracy (%)
Citation Faithfulness Rate (%)
Refusal Accuracy on Ungrounded (%)